

Experiment No: 6

Title:

Application of String Functions and Regular Expressions (Regex) for Email Validation, Data Extraction, and Text Analysis.

Software/Tools Required:

- Visual Studio Code
- Web Browser such as Google Chrome
- HTML, CSS, and JavaScript

Theory:

Strings are one of the most commonly used data types in JavaScript and are used to store and manipulate textual data. JavaScript provides a wide range of **string methods** that help perform operations such as searching, extracting, modifying, and analyzing text. Commonly used string functions

include `length`, `toUpperCase()`, `toLowerCase()`, `trim()`, `slice()`, `substring()`, `replace()`, `split()`, `includes()`, `startsWith()`, and `endsWith()`. These methods make text processing efficient and improve the overall functionality of web applications.

A **Regular Expression (Regex)** is a sequence of characters that defines a search pattern. In JavaScript, regular expressions are used to validate user input, search for specific patterns within text, replace matched content, and extract useful information from strings. Regex is widely used in form validation, password verification, email validation, and data processing because it allows complex pattern matching with minimal code.

By combining string manipulation techniques with regular expressions, JavaScript can efficiently process, validate, and analyze textual data. These concepts are widely used in modern web applications for user input validation, search functionality, data filtering, and text processing.

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Result/Conclusion:

Thus, JavaScript string functions and regular expressions were successfully implemented for email validation, data extraction, and text analysis. The application accurately validated user input, extracted required information, and analyzed textual data, demonstrating the practical use of string manipulation and pattern matching in web development.

